

SHIVANAND KUMAR

SENIOR DATA ENGINEER | AGENTIC AI & DATA PLATFORM AUTOMATION | SQL & DATABRICKS

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FREELANCE PROFILE

Senior Data Engineer with 5 years of experience delivering enterprise data platforms, analytics automation, and AI-driven engineering workflows across HP, Amazon, and BYJU'S. Supports SQL and Python automation, Databricks and PySpark pipelines, data quality and observability, platform migrations, agentic engineering workflows, and LLM-data evaluation. Combines hands-on delivery with stakeholder communication and measurable business outcomes.

CLIENT-READY CAPABILITIES

Data Engineering & Analytics: Advanced SQL, Python, PySpark, ETL/ELT, data modeling, pipeline development, KPI design, reporting automation, and data-quality checks

Data Platforms & Cloud: Databricks, Delta Lake, Unity Catalog, Spark, AWS S3, Glue, Redshift, lakehouse systems, migration, performance tuning, and observability

Agentic AI & Automation: Agentic workflows, GitHub Copilot agent skills, structured generation, deterministic validation, Microsoft Waza evaluation sets, human-review controls, and LLM evaluation

Delivery & Collaboration: Requirements analysis, technical documentation, stakeholder management, project ownership, operational reporting, and cross-functional coordination

PROFESSIONAL EXPERIENCE

HP | Research & Development Center | Bengaluru

Data Scientist II | Data Engineering & AI Platform Automation | Jun 2024 - Present

- Led end-to-end design and rollout of an **AI-driven data-product contract automation framework across 47 Databricks pipelines**, reducing manual documentation effort from days to hours.
- Owned automated monitoring across **54 production pipelines**, covering failure detection, signal prioritization, and corrective workflows; reduced recurring manual checks by **85-90%**.
- Led Splunk deprecation and logging standardization through a reusable migration and code-refactoring framework; migrated **40 production pipelines with zero migration-related runtime failures**, cutting manual effort by ~90%.
- Optimized firmware-log and Cloudflare telemetry pipelines through partitioning, distributed PySpark processing, join optimization, and caching, delivering a **~75% runtime reduction** while lowering compute overhead.
- Owned the **five-phase agentic generation and validation control loop** for configurations, schemas, data sources, target DDLs, and 20+ transformations; integrated Microsoft Waza evaluation sets into the Data Quality framework for repeatability and regression checks before human approval.
- Led Spark-based deduplication for an enterprise lakehouse dataset, reducing data volume from **21B to 400M rows** while improving downstream performance and infrastructure efficiency.
- Designed lakehouse observability for pipeline health, SLA tracking, proactive monitoring, and incident diagnosis, contributing to **~\$1.55M in annual infrastructure savings**.

Amazon | Oct 2022 - Jun 2024

Business Analyst | Jul 2023 - Jun 2024 | Bengaluru

- Owned analytics for supply-chain systems supporting **15M+ users**; built production SQL pipelines and BI datasets for operational dashboards and leadership reporting.
- Automated recurring operational metric workflows using SQL and Python, reducing manual reporting effort by **~70%**.
- Designed complex SQL transformations that delivered a **~30% runtime reduction**.

Recognition: Director-level "Minesweeper" and "Automation Expert" awards for automation impact.

Machine Learning Data Associate II | Oct 2022 - Jul 2023 | Hyderabad

- Curated and quality-checked supervised fine-tuning datasets for LLMs through annotation, validation, and dataset-quality workflows.
- Evaluated prompt-response pairs and curated training data to improve model consistency and support LLM training and evaluation cycles.

BYJU'S | Oct 2021 - Oct 2022 | Lucknow

Centre Head - Operations

- Led a **16-member cross-functional team** across centre operations, scheduling, delivery, and performance management.
- Used SQL dashboards and data-led operating reviews to drive **~30% quarterly growth** and achieve 100% centre performance ratings across North India.

EDUCATION & CERTIFICATIONS

M.Tech, AI & Data Science Engineering | Indian Institute of Technology (IIT) Patna | Jul 2026 - Present

B.E. in Mechanical Engineering | University Institute of Engineering and Technology (UIET), Panjab University, Chandigarh | 2016 - 2020

Certifications: DeepLearning.AI - Supervised Machine Learning | Google - AI Essentials & Data Analytics | HackerRank - SQL (Advanced)